

Started on	Saturday, 1 August 2026, 9:37 AM
State	Finished
Completed on	Saturday, 1 August 2026, 11:17 AM
Time taken	1 hour 40 mins
Grade	100.00 out of 100.00

Question 1

Correct

Mark 100.00 out of 100.00

Use Gaussian elimination without partial pivoting to solve a matrix.

Hint: First value is the number of unknowns, remaining values are the elements of the matrix.

For example:

Input	Result
3	X0 = 53.35 X1 = -8.88 X2 = -4.40
1	
2	
4	
18	
2	
12	
-2	
9	
5	
26	
5	
14	

Answer: (penalty regime: 0 %)

Reset answer

```

1  '''Program to solve a matrix using Gaussian elimination without partial pivoting.
2  Developed by: VEDHA M
3  RegisterNumber: 21225230292
4  '''
5  import os
6  os.environ["OPENBLAS_NUM_THREADS"]="1"
7  import numpy as np
8  import sys
9  n = int(input())
10 a = np.zeros((n,n+1))
11 x = np.zeros(n)
12 for i in range(n):
13     for j in range(n+1):
14         a[i][j]=float(input())
15 for i in range(n):
16     if a[i][i] ==0.0:
17         sys.exit('Divide by zero detected!')
18     for j in range(i+1,n):
19         ratio=a[j][i]/a[i][i]
20         for k in range(n+1):
21             a[j][k]=a[j][k] -ratio*a[i][k]
22 x[n-1]=a[n-1][n]/a[n-1][n-1]
23 for i in range(n-2,-1,-1):
24     x[i]=a[i][n]
25     for j in range(i+1,n):
26         x[i]=a[i][n]
27     for j in range(i+1,n):
28         x[i]=x[i]-a[i][j]*x[j]
29     x[i]=x[i]/a[i][i]
30 for i in range(n):
31     print('X%d = %0.2f'%(i,x[i]),end=" ")

```

	Input	Expected	Got	
✓	3 1 2 4 18 2 12 -2 9 5 26 5 14	X0 = 53.35 X1 = -8.88 X2 = -4.40	X0 = 53.35 X1 = -8.88 X2 = -4.40	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 100.00/100.00.